Task 3 EEG forecasting and nominal charge

E017 research note Satyam Awasthi 9 November 2026

Transfer to held-out Task 3 trials. Both models forecast new trials from the same participants, waveform types and amplitude tiers, with post-trigger R^2 above 0.9979 at all three target widths. The pooled waveform model gave lower forecast error at every width.

Models

For each participant, electrode and target window, a linear regression predicted the next mean EEG voltage from three consecutive 50 ms means at the same electrode. For a target starting at 200 ms, for example, the histories covered 50–100, 100–150 and 150–200 ms. Both forecasting approaches used

$$\widehat{y} = \alpha + b_1 h_1 + b_2 h_2 + b_3 h_3 + \beta Q,$$

where h_1, h_2, h_3 are the three history means and Q is nominal cathodic charge in μC. The history-only fit omitted the βQ term. Nominal charge was the recorded peak current times the 0.4 ms cathodic phase duration and the ideal waveform area factor. EEG was scalp-average referenced without trial-baseline subtraction.

Pooled waveform model

One set of coefficients was fitted across all eligible waveform types within each participant, electrode and target window.

Waveform-specific model

A separate set of coefficients was fitted within each known waveform type, for the same participant, electrode and target window. Forecasting therefore used the waveform identity to choose the fit.

Held-out test

The analysis used 1,382 trials from 10 participants and all 62 scalp electrodes. Each participant–waveform series was split into five chronological chunks. Each chunk was held out once, with all three amplitude tiers represented in training. Both approaches used identical trial splits. All target widths began at 200 and 300 ms post-trigger, with matched starts at −350 and −250 ms before the trigger. The two 200 ms targets overlap by 100 ms.

Forecast accuracy on held-out Task 3 trials

		Post MAE (μV)		Post held-out R^2		Error reduction from charge	
Target	Model	History	+ Charge	History	+ Charge	Post (95% interval)	Pre
50 ms	Pooled	2.198	2.195	0.999699	0.999699	+0.21% (−1.03% to +1.62%)	−0.97%
50 ms	Waveform-specific	2.454	2.514	0.999504	0.999464	−3.50% (−6.14% to −0.93%)	−4.84%
100 ms	Pooled	2.297	2.295	0.999532	0.999532	+0.29% (−0.92% to +1.79%)	−0.78%
100 ms	Waveform-specific	2.613	2.699	0.999062	0.998965	−4.25% (−7.03% to −1.66%)	−4.63%
200 ms	Pooled	2.906	2.872	0.999034	0.999092	+0.62% (−1.04% to +2.81%)	−0.98%
200 ms	Waveform-specific	3.293	3.389	0.997973	0.997923	−3.60% (−5.15% to −2.00%)	−5.76%

Positive error reduction means lower held-out mean squared error (MSE) than the corresponding history-only fit. Scores weight waveform–tier cells equally within participants, then weight participants equally. Intervals use 10,000 participant bootstrap draws.

The high held-out R^2 chiefly reflects how similar adjacent voltage windows are. For the pooled forecast, the interval for charge-related improvement included zero at all three widths. Adding charge increased error in the waveform-specific fits. Task 4 transfer of the charge slope remains untested.

Waveform results and Task 4 plan

Results by waveform at 200 ms

Both columns of MAE use the history-plus-charge model. Each waveform row averages the participants with eligible trials for that waveform.

		Post MAE (μV)		Post error reduction from charge	
Waveform	Participants	Pooled	Waveform-specific	Pooled	Waveform-specific
Sine	9	3.128	3.605	+0.73%	−5.28%
Square	10	3.036	3.627	+0.54%	−3.60%
Triangle	10	2.585	2.933	+0.32%	−5.89%
Linear decrease	9	2.825	3.659	+0.08%	−3.31%
Linear increase	9	2.822	3.175	+0.05%	−1.40%

Charge calibration without EEG history

A separate calibration fit used waveform intercepts and either a shared slope or one slope per waveform. This estimates the dose association without conditioning on recent post-trigger EEG, which may already carry a dose-related response.

200 ms calibration model	Post MAE (μV)	Error reduction vs. mean (95% interval)
Waveform mean	58.85	Reference
Shared charge slope	59.38	−1.61% (−2.59% to −0.82%)
Separate charge slopes	60.20	−4.39% (−7.28% to −1.73%)

Proposed charge adjustment in Task 4

1. **Match the features and calibrate on Task 3.** Use the same reference, electrode set and consecutive time windows in both tasks, for example five 200 ms means per electrode over 50–1,050 ms. Estimate one shared charge slope $\widehat{\beta}^{(3)}_{s,e,k}$ for each calibrated participant s, electrode e, and window k, using within-waveform dose variation. This requires extending Task 3 extraction through 1,050 ms and checking whether the slopes predict held-out dose-related variation. Current calibration covers targets only through 500 ms.

2. **Adjust the observed Task 4 features.** Estimate cathodic charge Q_i for each Task 4 trial i by integrating the aligned current-monitor trace over the cathodic phase. Use the same charge definition in Task 3 and check that Task 4 values lie within the calibration range. Choose a common reference charge $Q_{\mathrm{ref},s}$ per participant from Task 3 calibration and freeze it for every waveform:

$$x^{\mathrm{adj}}_{i,e,k} = x^{(4)}_{i,e,k} - \widehat{\beta}^{(3)}_{s,e,k}\big(Q_i - Q_{\mathrm{ref},s}\big).$$

   For example, a slope of $20\,\mu\mathrm{V}/\mu\mathrm{C}$ and a trial charge $0.05\,\mu\mathrm{C}$ above the reference give a $1\,\mu\mathrm{V}$ subtraction from that window mean. The shared slope and independently measured charge allow this operation without the held-out Task 4 waveform label.

3. **Evaluate the adjustment.** Compare waveform decoding from original and adjusted features on the same comparison-held-out splits, keeping the pre-trigger control and reporting the two presentation positions separately. The current results give weak support for a transferable correction. Task 4 testing would quantify what changes after removing the estimated linear dose component.

Analysis record `Docs/E017_TASK3_CHARGE_RESULTS.md`